Course Project Report

An Efficient Truss Bridge

MCAE 200 – Statics

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Experiment Conducted on: 20/11/2025

Report Submitted on: 28/11/2025

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Objectives

The purpose of this report is to explain the educational value of truss-bridge analysis, outline the theoretical background and design constraints of the project, and present the methods, structural decisions, and performance outcomes that resulted in the final optimized bridge.

Introduction

Statics is a branch of mechanics that studies equilibrium, or the conditions under which structures remain at rest under specific loads. The goal of static analysis is to design elements and systems that safely and efficiently withstand operational loads over long periods.

There are several types of structural systems, each responding differently to applied loads. During the Statics course, two basic types were studied: frame and truss structures. Their key difference lies in the distribution of internal forces. According to Newton's third law, for every force there is an equal and opposite reaction. This principle provides a clear explanation for the efficiency of trusses. In trusses, the elements are connected by pinned joints that allow rotation at the joints. Due to these rotations, when a load is applied to a truss joint, the forces are transmitted through the elements as axial forces, causing tension or compression. The counterforces arising at the joints balance the applied loads, and bending moments are negligible, at least in theory. This property allows the material to be used predominantly in its maximum strength state, whether tension or compression, reducing the amount of material required.

Frame structures, on the other hand, behave differently because their joints are rigid. When loads are applied to beams, bending moments and shear forces are generated, causing deformation of the entire structure. This requires the use of stronger materials and more rigid connections, which complicates the design and calculation of endurance. Despite this, frame systems are extremely useful where stable, massive, and durable support is required. Multi-story buildings, office towers, and bridges often use frame structures to safely distribute both vertical and horizontal loads.

Trusses, in contrast, are ideal for situations where weight reduction and material efficiency are important, such as roofs, lightweight bridges, structural furniture components,

or temporary infrastructure. In practice, frames and trusses rarely exist separately; they typically work together as parts of a single structural system. The frame provides overall rigidity and stability, while the truss efficiently distributes loads through axial action.

Trusses are particularly suitable for educational purposes, and constructing a truss bridge is a common final project in Statics for several key reasons. First, their two-dimensional representation contains all the necessary structural information for further analysis and design. Second, the triangular geometry of trusses is inherently rigid, making these systems stable and predictable. Finally, the internal forces in truss elements can be determined using simple, proven manual methods such as the Method of Joints and the Method of Sections. This makes truss structures accessible to students and an effective means of demonstrating the fundamental principles of statics.

In statically determinate trusses, the relationship between elements, joints, and reactions is crucial. A common check is the formula: $m + 3 = 2j$, where m is the number of members and j is the number of joints. If this condition is met, the internal forces can be determined solely from the equilibrium equations, neglecting element deformations. If a truss has too many members, it becomes statically indeterminate, requiring additional methods to calculate the forces in individual members, which complicates design and analysis.

Real-world bridges require careful design, as their primary function is to transfer loads from one side of the span to the other. Loads are distributed throughout the entire structure, with each joint gradually subjected to forces as traffic moves across the span. However, in the classroom project, the weight was suspended at the center of the span. This design required a different approach, as the goal was to efficiently distribute the load and create a lightweight structure.

Without knowing the specific material qualities, engineers cannot design the bridge, as the strength and stiffness of each member determine whether it can safely carry the tension or compression assigned to it. For example, cold-formed steel generally has a higher strength-to-weight ratio than wood, meaning it can support larger loads with less material. Steel also maintains significant strength after local buckling, resists fire better, and is more dimensionally stable under environmental changes. In contrast, wood is more variable: while it can perform well in tension, it is more prone to buckling under compression, and its strength can be reduced by natural defects such as knots or grain variations. These differences affect how structural members must be designed and dimensioned to carry loads safely and efficiently (Scottsdale Steel Frames, 2025).

For the classroom project, students were required to use popsicle sticks as the structural members, with pin-like joints made from materials such as wires, nails, or small bolts. Popsicle sticks behave more like timber: they are relatively strong in tension but weaker in compression. Understanding how loads distribute through the truss and designing members to resist both tension and compression is crucial, even in this simplified model.

Truss Analysis and Design

When designing a popsicle bridge, it is crucial to first understand the limitations of the material and design the structure accordingly. Due to their flat and slender shape, popsicle sticks are prone to bending and buckling under compressive forces. From my lab experiments, I observed that the longer a member is, the more it is prone to bending and breaking at the points of connection or glue. This property was carefully considered when designing the truss, ensuring that compression members remained relatively short and that there were fewer of them overall. Another important constraint was the overall length of each member. According to the project requirements, the bridge span was limited to four popsicle sticks in length. Members longer than this had to be joined using glue or cut if necessary, introducing potential weak points under both tension and compression. Therefore, it was crucial to minimize member length to no more than two sticks in a row, limiting the number of cuts and glue joints to at most one per member.

The loading condition for this project was simplified: a single concentrated load was applied at the center of the span. The bridge design had to distribute this load efficiently, minimizing compression in the members while keeping their lengths within acceptable limits. By applying the weight only on the bottom central joint and testing multiple examples and designs using the Truss Simulator for JHU Engineering Innovation platform, I identified the optimal design for my competition. I followed a strategy of testing as many configurations as possible, including ideas from brainstorming and online examples. Each design was built in the simulator, the load was applied, and the stress on each member and its length were recorded. Members experiencing excessive load were either shortened or reinforced by combining multiple sticks, depending on the stress they experienced. For example, members bearing more than 80% of the maximum load were reinforced by tripling their length, whereas members carrying more than 30% of the maximum load were doubled in length.

To evaluate different designs, several configurations were analyzed. The first alternative design (Figure 1) resulted in a total member length of approximately 161 cm. No individual member exceeded 80% of the maximum applied load, which initially seemed promising. However, seven members were under compression and eight under tension, and all of these members exceeded the length of a single popsicle stick, requiring joining. Combined with the overall weight and height of the structure, this design created potential instability, making it less practical given the material constraints.

The second alternative design (Figure 2) had fewer members under compression – six – and tension – seven – and the total member length was reduced to around 140 cm. All members except the central joint stayed below 60% of the maximum load, which improved efficiency. Despite these advantages, most members still required joining, which could reduce structural integrity based on experimental observations with popsicle sticks.

The final competitive design (Figure 3) minimized total member length to approximately 113 cm. Only two bottom members reached 67% of the maximum load, with two additional members approaching this threshold. Importantly, only five members needed to be joined, significantly fewer than in the previous alternatives. Two sticks in compression in the middle experienced very low stress, around 30% of the maximum load, which allowed for less material use in the construction phase. This combination of optimized load distribution, minimized joins, and reduced total length made it the best candidate for construction.

Manual calculations of member forces were then performed using both the Method of Joints and the Method of Sections (Figures 4a–b). These step-by-step calculations illustrated how loads were transferred through the truss and confirmed that the chosen design could safely carry the applied load while respecting material limitations.

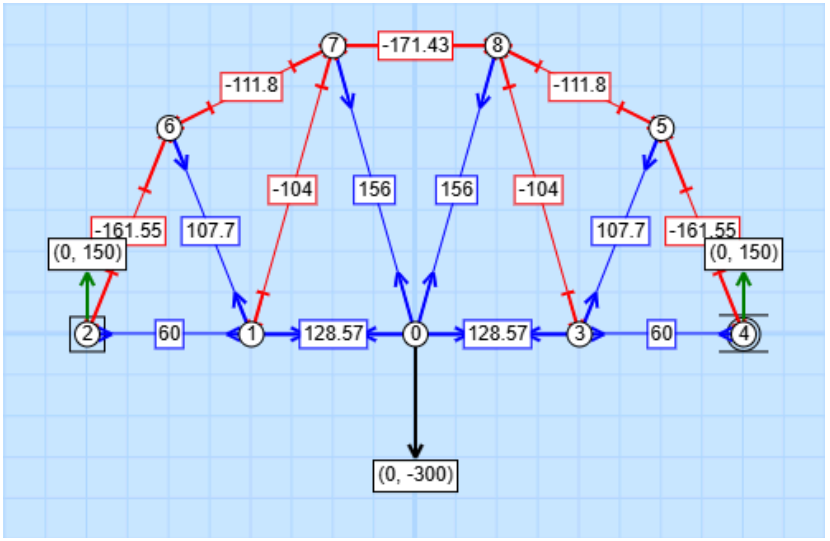


Figure 1. First alternative design: total member length ~161 cm

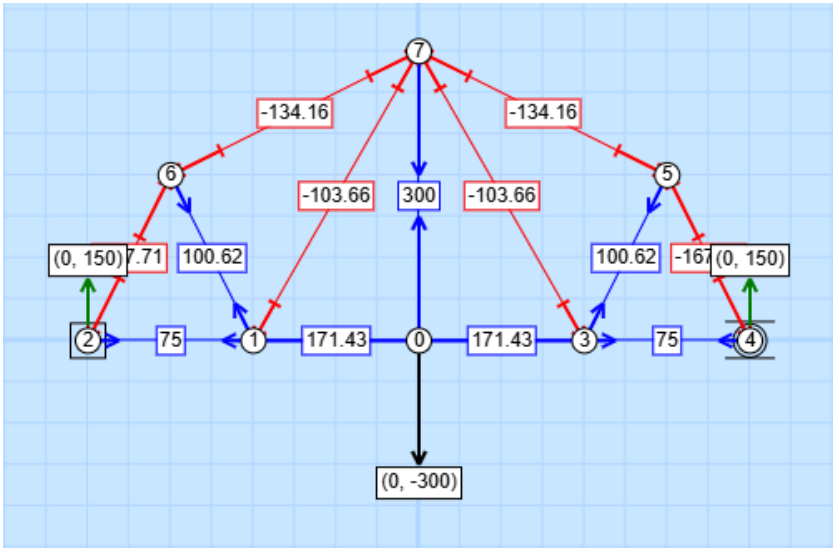


Figure 2. Second alternative design: total member length ~140 cm

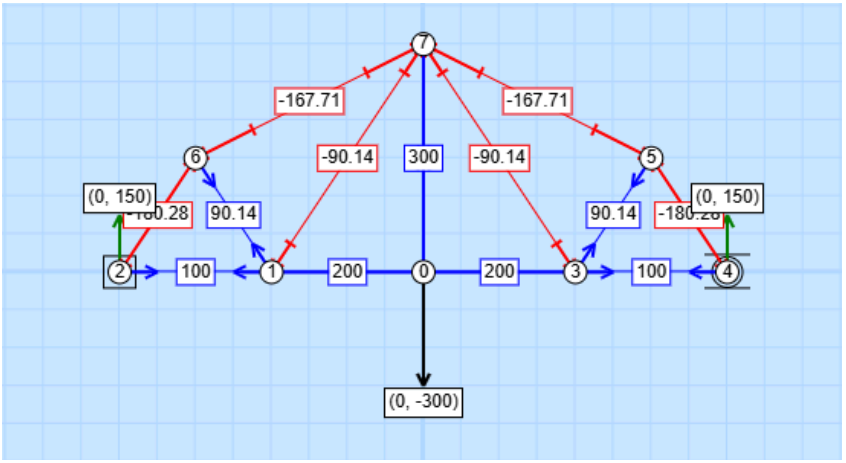


Figure 3. Final competitive design: total member length ~113 cm

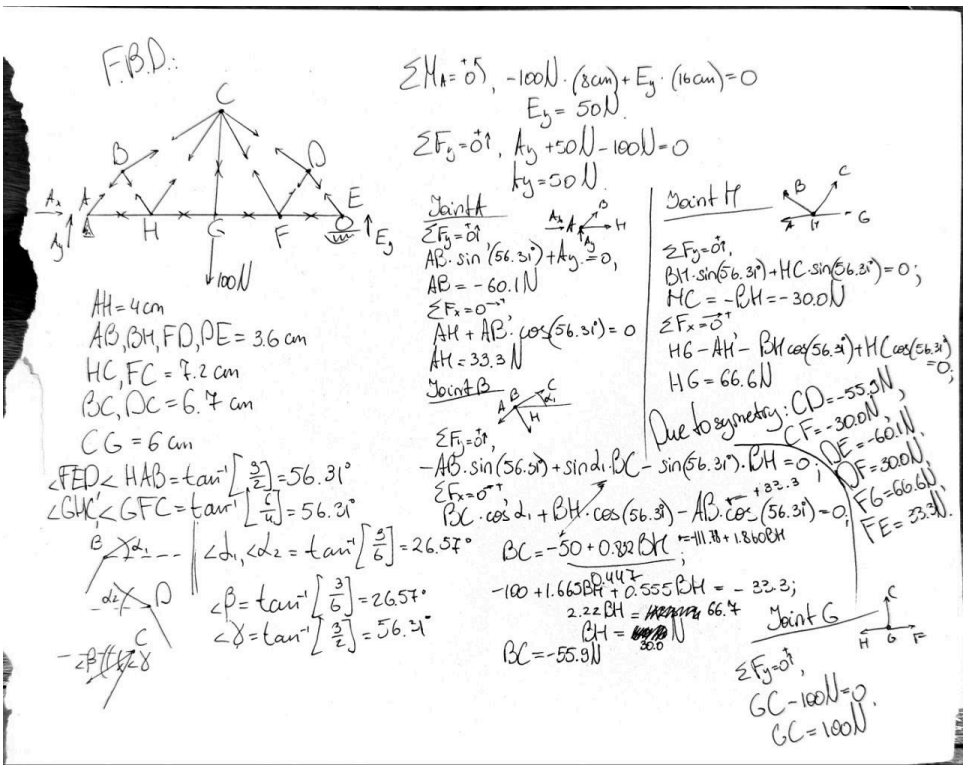


Figure 4a. Method of Joints calculation

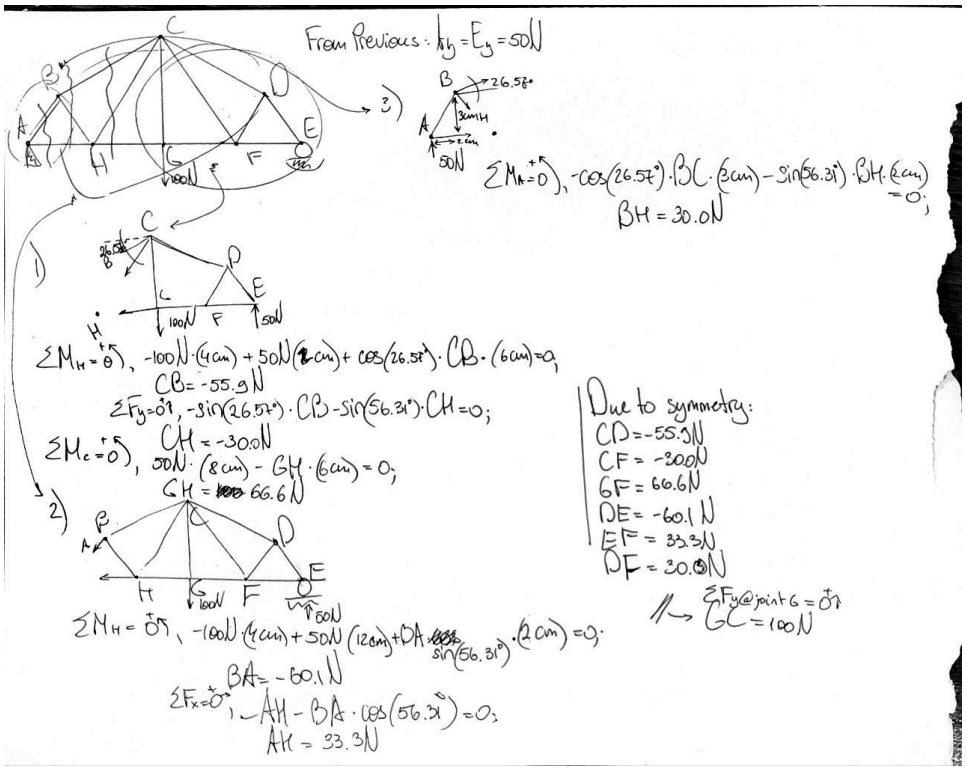


Figure 4b. Method of Sections calculation

Physical Model

The next step in the project involved determining the most effective way to assemble and join the popsicle sticks, ensuring that both tension and compression forces were managed efficiently. Using these observations and calculations, the final bridge was constructed with minimal joins, optimized member lengths, and careful attention to the distribution of compressive and tensile forces throughout the truss.

To choose the correct joining technique, I first conducted several small-scale experiments with individual popsicle sticks under tension. I drilled holes for a small bolt, inserted it, and applied force to a single stick. In every case, the member failed at the joint: the wood around the hole tore out together with its grain, causing complete joint failure. Increasing the number of glued sticks from one to two did not help – the same failure occurred as soon as the tensile force increased.

Through these tests I learned a critical principle: during tension, it is essential to maximize the amount of wood surrounding the bolt hole. Any insufficient cross-section caused immediate failure. Based on this, I decided that every tension member must consist of at least three layers of popsicle sticks, specifically at the joint locations. To strengthen these joints, I added two small 2–3 cm wooden plates (cut from popsicle sticks' ends) on both sides of the main member. After gluing the three-stick member together, I bonded these reinforcement plates on both sides, drilled through all layers, and installed the bolt. Experiments showed that this layered configuration with PVA glue could withstand extremely large loads – significantly more than my own body weight – so I selected it as the primary method for all high-tension members.

The central bottom member and joint, which carries nearly 100% of the overall load, was constructed using this technique, but with even more reinforcement (Figure 5a). Members that carried around 70–80% of the load received the same design, though with fewer layers (three instead of five). These decisions directly matched the results of my simulator stress analysis, and the later physical failure pattern confirmed the calculations.

Compression members presented a completely different challenge. The outer top chords of the bridge – large members located on the outside faces of the structure – were originally made by simply gluing two sticks together into a long beam, and adding two more to the sides of the cut. However, this failed during testing due to insufficient moment of inertia. Because popsicle sticks are thin and flat, they do not resist twisting under

compression load. The lack of thickness made the members extremely easy to rotate about their longitudinal axis.

To solve this, I redesigned the compression members into an I-shaped beam, inspired by the steel I-beams used in real construction (Figure 5b). Each long top chord consisted of four small stick segments forming two parallel layers. These two layers created enough width for me to glue an additional popsicle stick vertically on each side, forming flanges—exactly like an I-beam profile. Once assembled, this wooden I-section resisted rotation extremely well and withstood significantly higher compression forces during laboratory tests. This redesign eliminated the failure modes seen in earlier trials.

All gluing was performed using large metal vises, which ensured perfectly straight, uniform pressure while bonding. I used PVA glue for most members because it dries faster, but for the central high-load beam, I applied epoxy resin. The drawback was that epoxy requires a full day to cure, so fabrication time increased substantially, limiting how many epoxy-reinforced pieces I could produce.

Another practical difficulty was finding hardware suitable for the thick, multi-layered joints. Many of my reinforced members became very wide due to the added layers, and standard bolts were not long enough. After searching through available materials, I ended up using a long, thick wood screw instead of a bolt. I drilled the joint, inserted the screw, and cut off the excess length with an angle grinder. This allowed me to create reliable pin joints even with very thick member stacks.

Once both sides of the truss (the two “walls” of the bridge) were completed, the final assembly involved joining them together. I added several lateral braces primarily at the top of the structure, since most stability was needed there. Only a small number of braces were added along the bottom, just enough to prevent the two sides from spreading apart or collapsing inward. I intentionally left the center clear to serve as a “walkway” or roadway, avoiding unnecessary members across the mid-span. All joints were constructed as pin joints, capable of rotation; however, once the full structure was assembled, the geometry became fixed and the bridge no longer rotated.

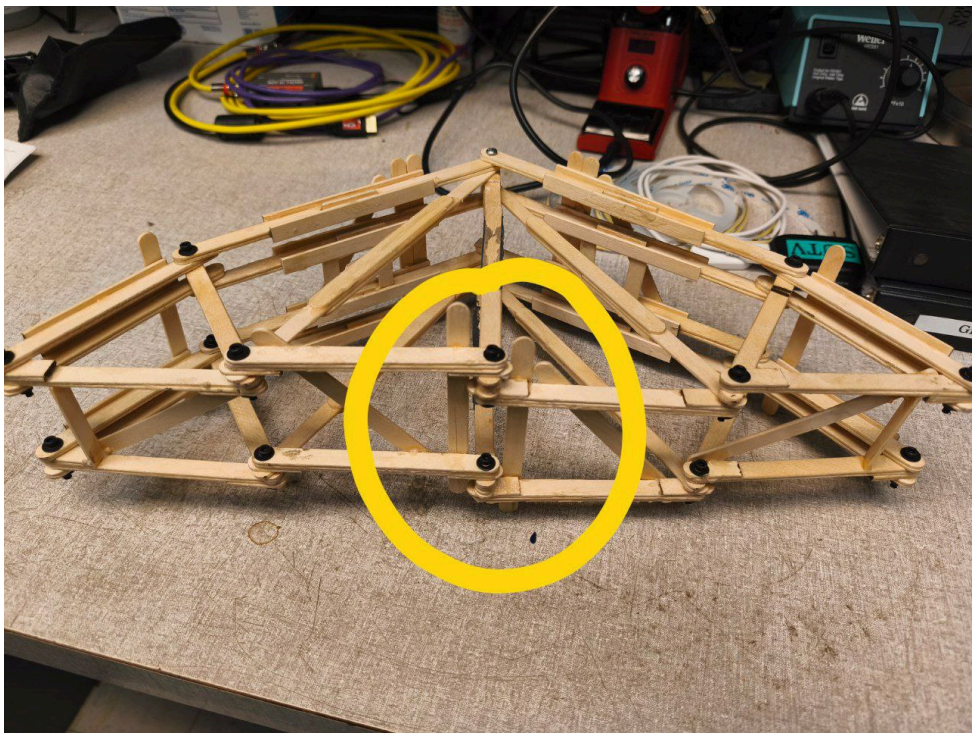


Figure 5a. Layering technique for critical members

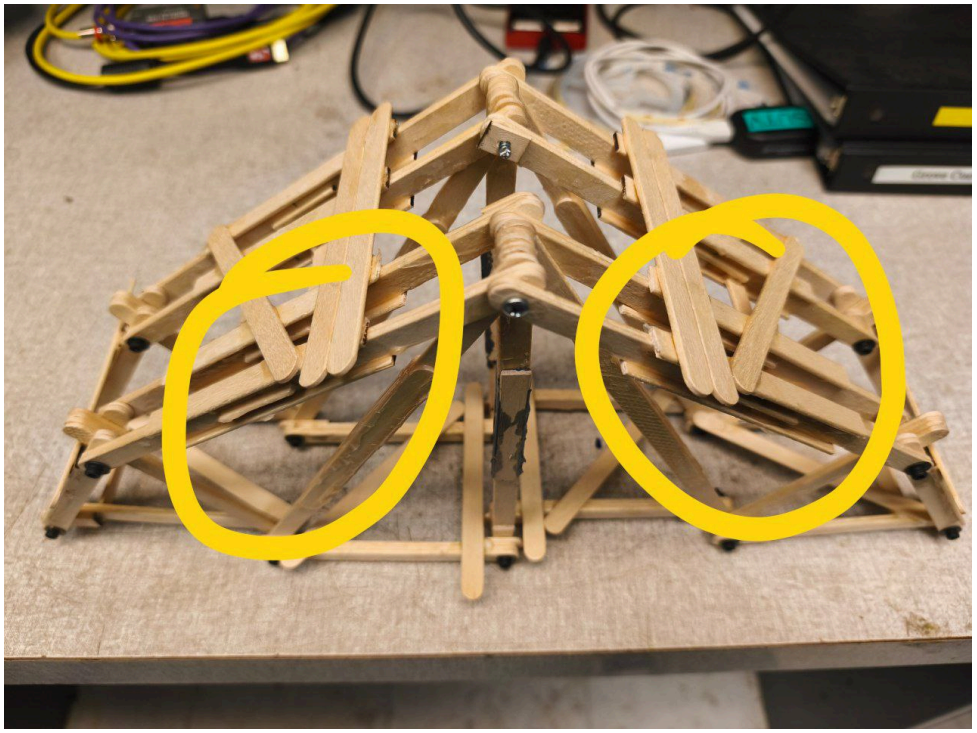


Figure 5b. Image of the I-shaped member

Conclusions

The results of this project demonstrate that the bridge design performed as intended, with failure occurring not in the critical load-bearing members themselves but at the central bottom joint connecting the lower chord to the side member (Figure 6). This joint, reinforced with three layers of wood, ultimately detached under load as the glue bond failed, causing the structure to collapse. Before failure, the bridge successfully supported 72.35 kg while weighing only 245 g, resulting in a load-to-weight ratio of 295.3, which verifies the effectiveness of the design strategy.

The performance confirms that the analytical approach – prioritizing compression resistance, adjusting member lengths based on simulated stress data, and reinforcing the most highly loaded elements – was fundamentally sound. The experiment also highlighted the limitations of material imperfections and adhesive strength, showing that structural failure was driven not by member capacity but by joint bonding reliability. Overall, this project demonstrated the value of truss bridges as an educational model for understanding load paths, stress distribution, and engineering decision-making. It also reinforced how simulation, design iteration, and real-world testing work together in practical structural engineering.

Figure 6. Failure location

Recommendations

Several improvements could substantially increase the structure's performance in future iterations. The most significant factor limiting the load-to-weight ratio was the adhesive: only two key extension-joint connections in the central bottom members were reinforced with epoxy, which proved dramatically stronger than PVA glue. Using epoxy in all joints would likely have prevented the failure entirely and allowed for a much lighter configuration, as fewer reinforcement layers would be required. This change alone could reasonably push the ratio well above 400.

Additional improvements could include performing multiple prototype tests to identify early failure points, followed by systematic redesign of joint layering, member geometry, and glue use. Certain I-shaped members might be simplified into regular beams if testing shows that they do not experience critical stresses, reducing overall weight. With more time, experimenting with alternative reinforcement patterns, re-evaluating compression zones, and optimizing the balance between stiffness and weight would further enhance structural efficiency.

References

Scottsdale Steel Frames. (2025, February 24). *Comparing steel vs wood framing: Why steel is better*. Scottsdale Steel Frames. https://www.scottsdalesteelframes.com/building-with-steel/comparing-steel-vs-wood-framing-why-steel-is-better

Appendix

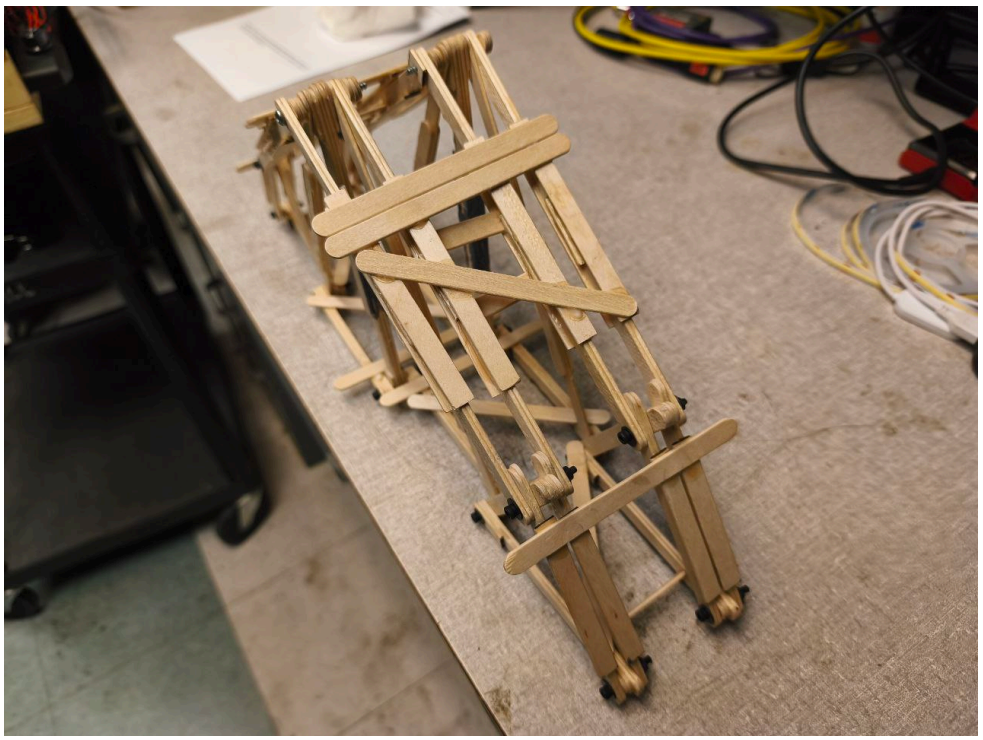


Figure 7. Side view of the final bridge

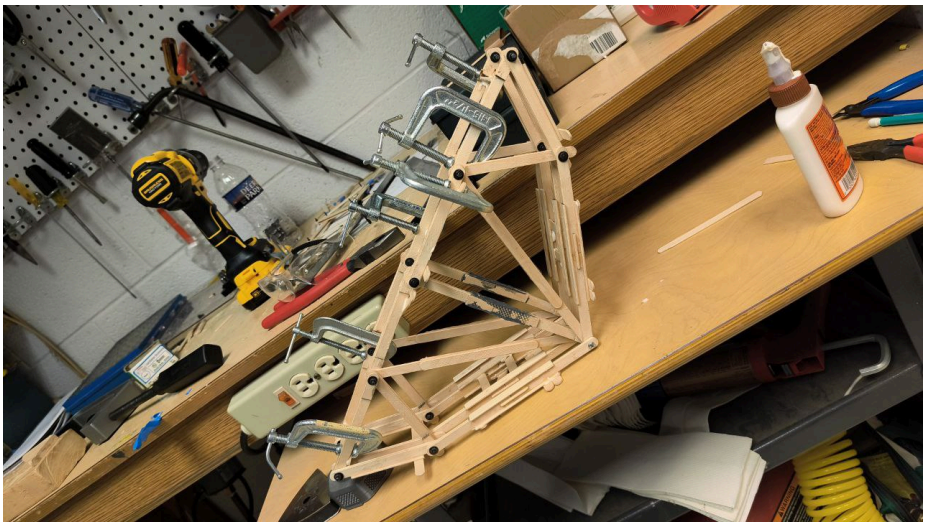


Figure 8. Gluing process of the bridge frame

Figure 9. Individual I-shaped members before assembly

Figure 10. Gluing process of I-shaped members

Figure 11. Five-layer epoxy-reinforced central member during assembly

Figure 12. Compression test failure of early prototypes

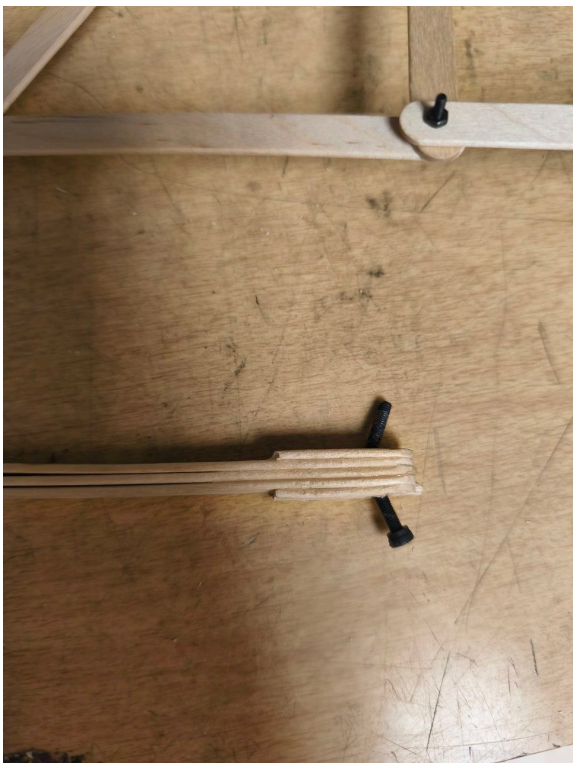


Figure 13. Successful test of the five-layer central member (bent bolt)

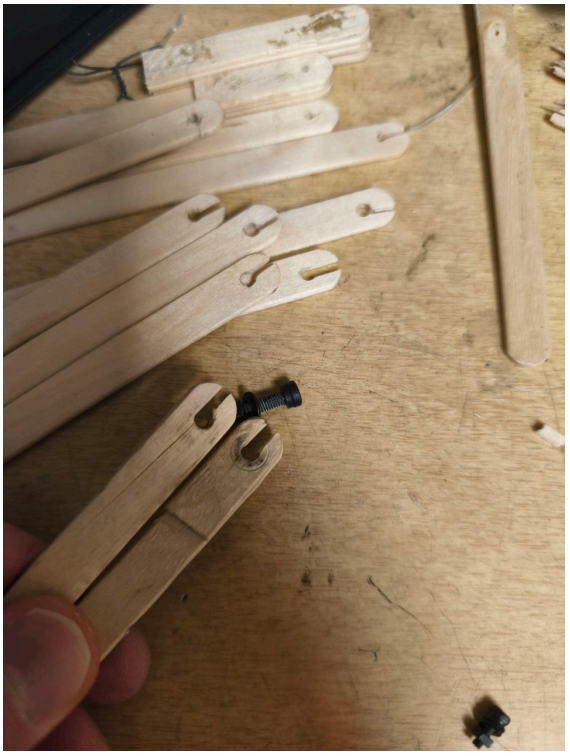
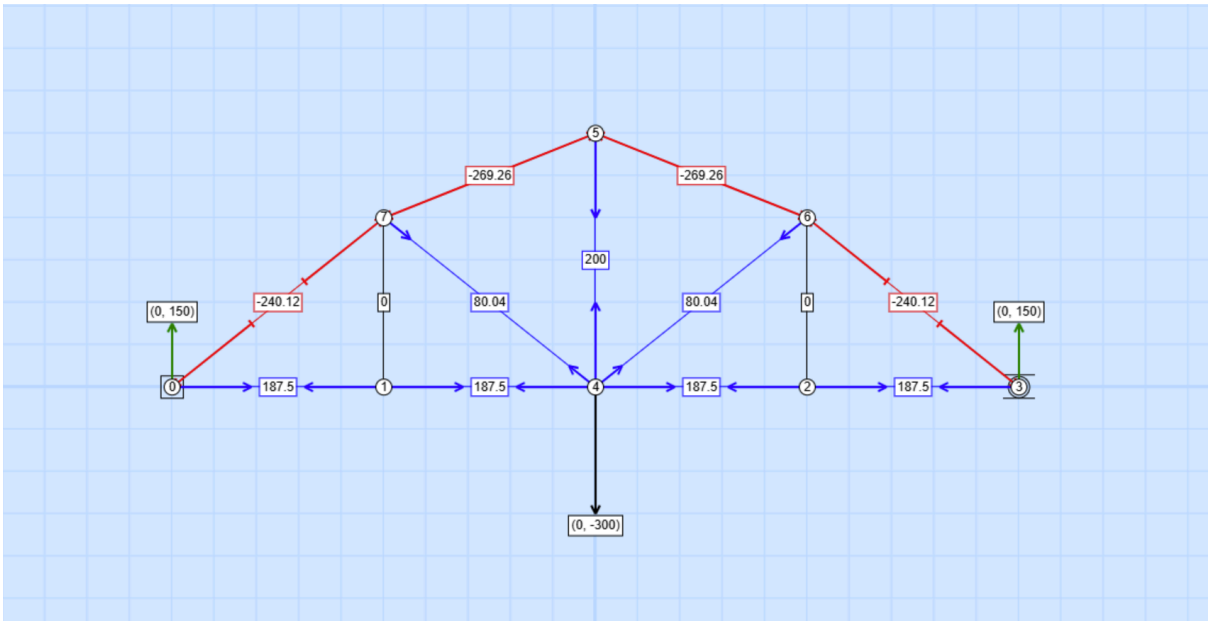


Figure 14. Multiple failures during tension tests (joint failures)

Figure 15. Additional simulation results from early design iterations (labeled a–j):



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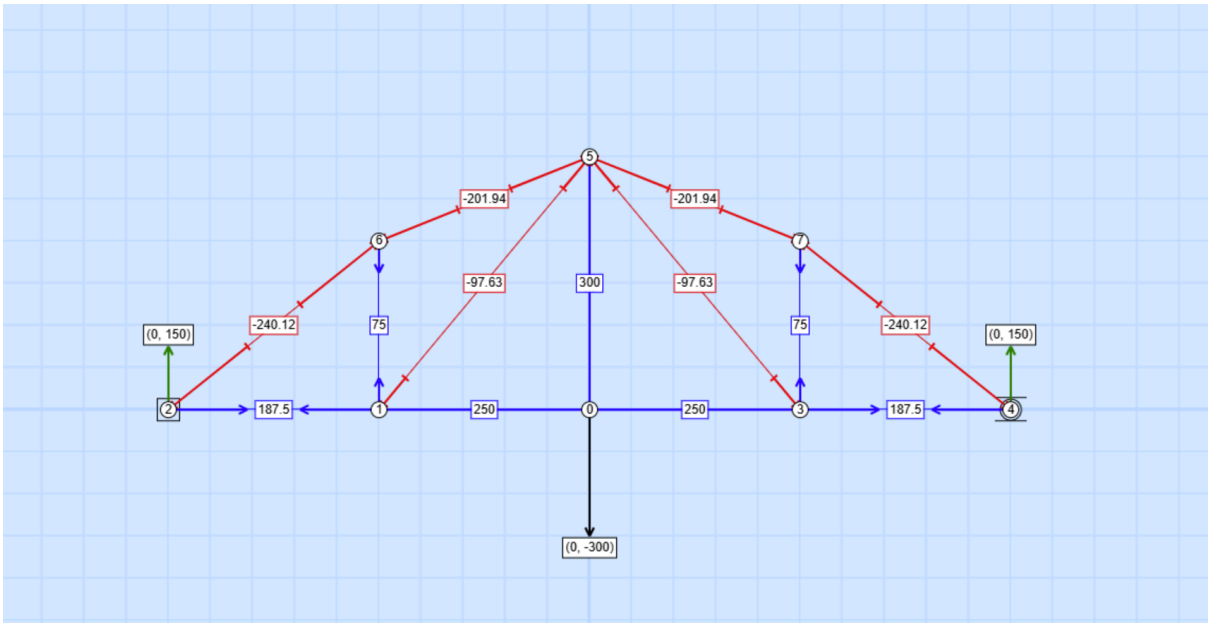

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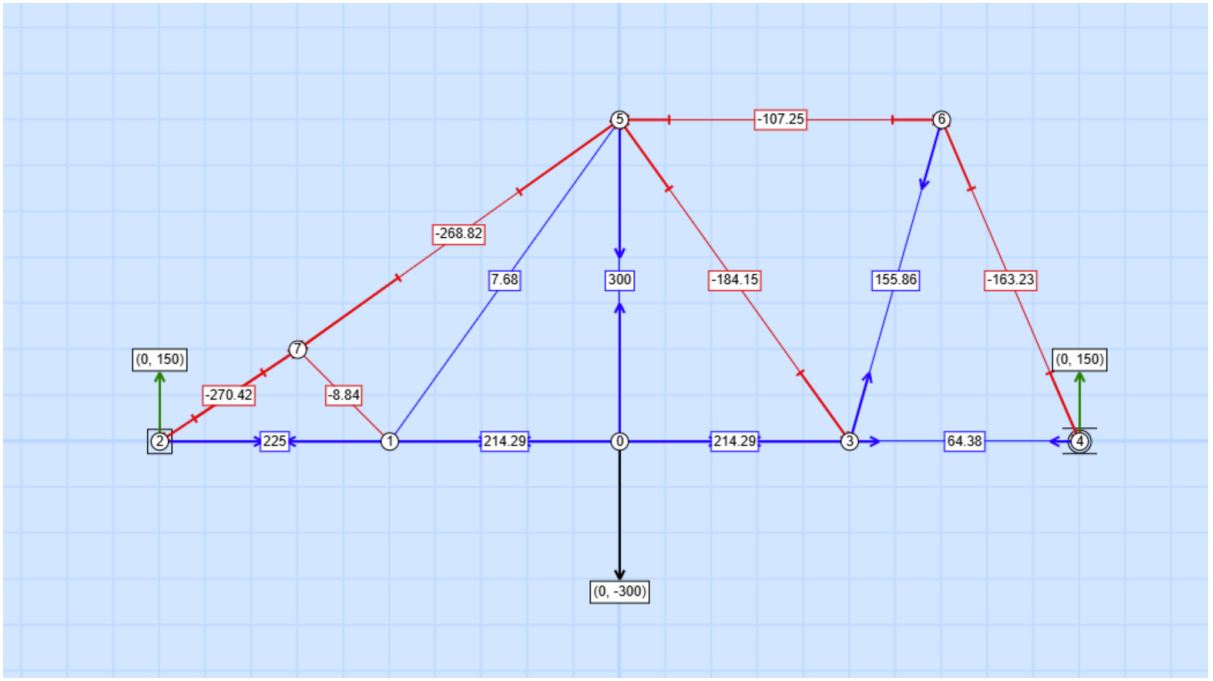
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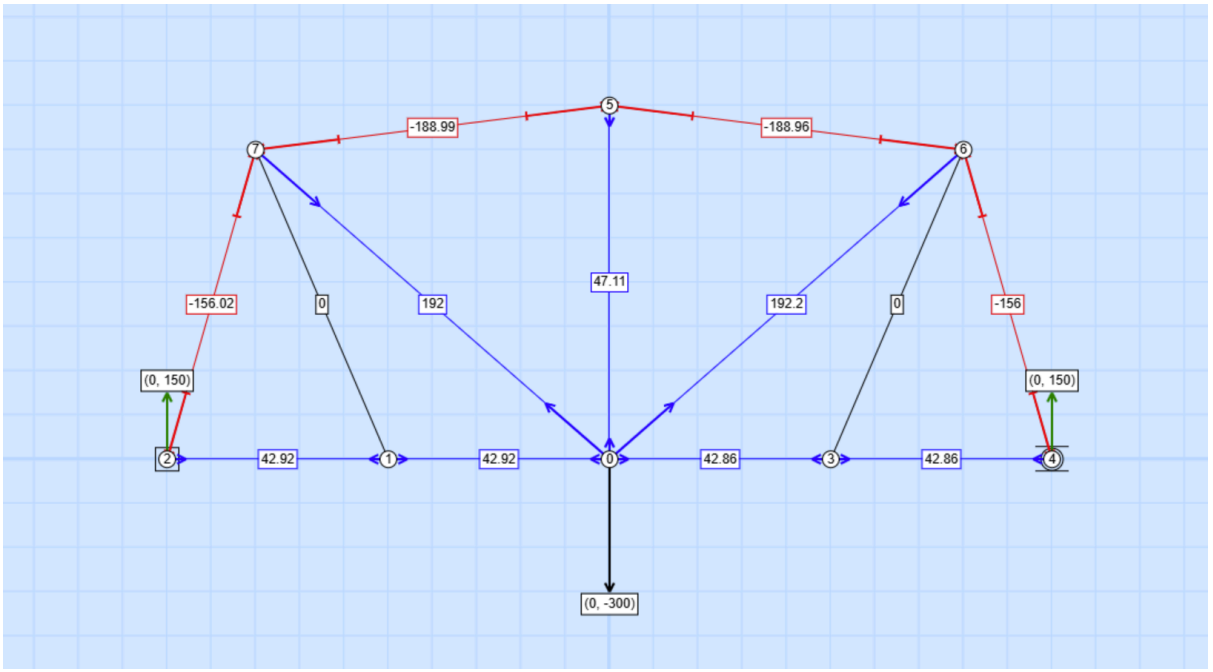
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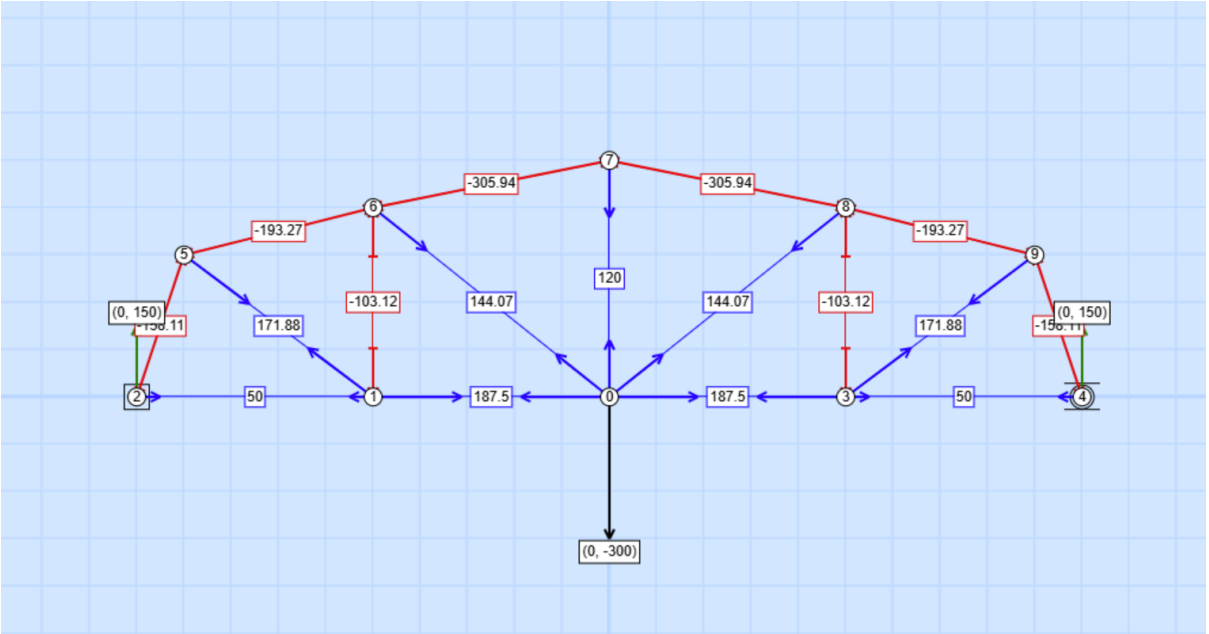
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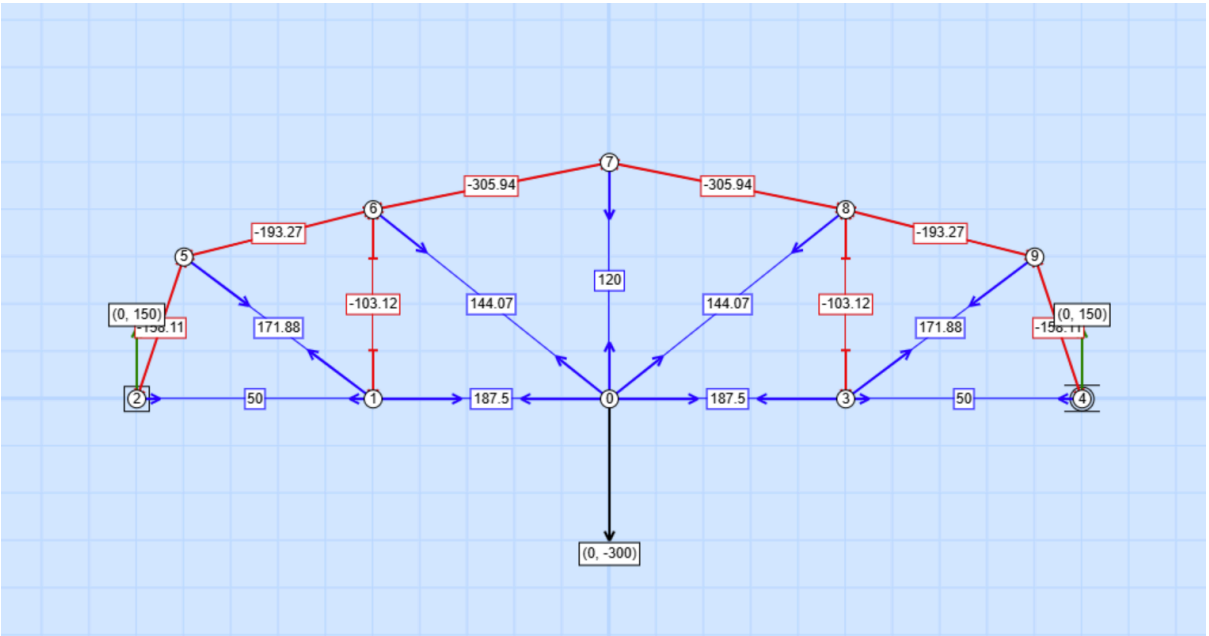
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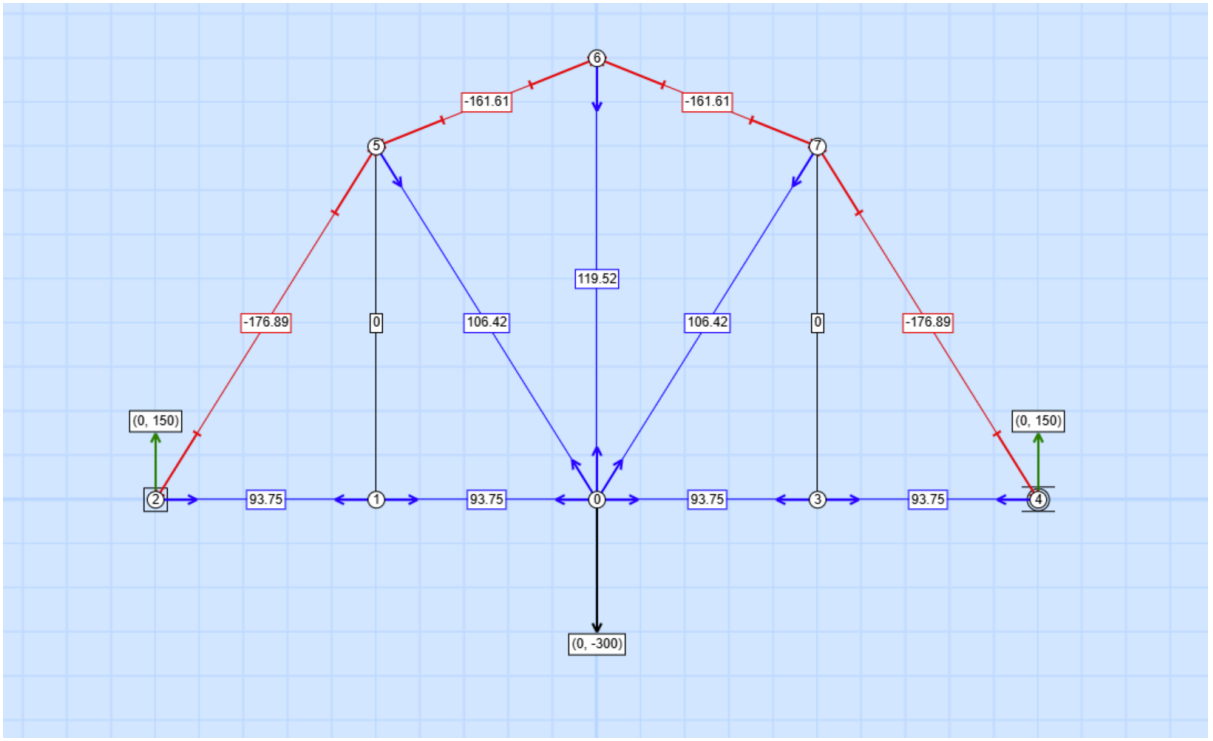
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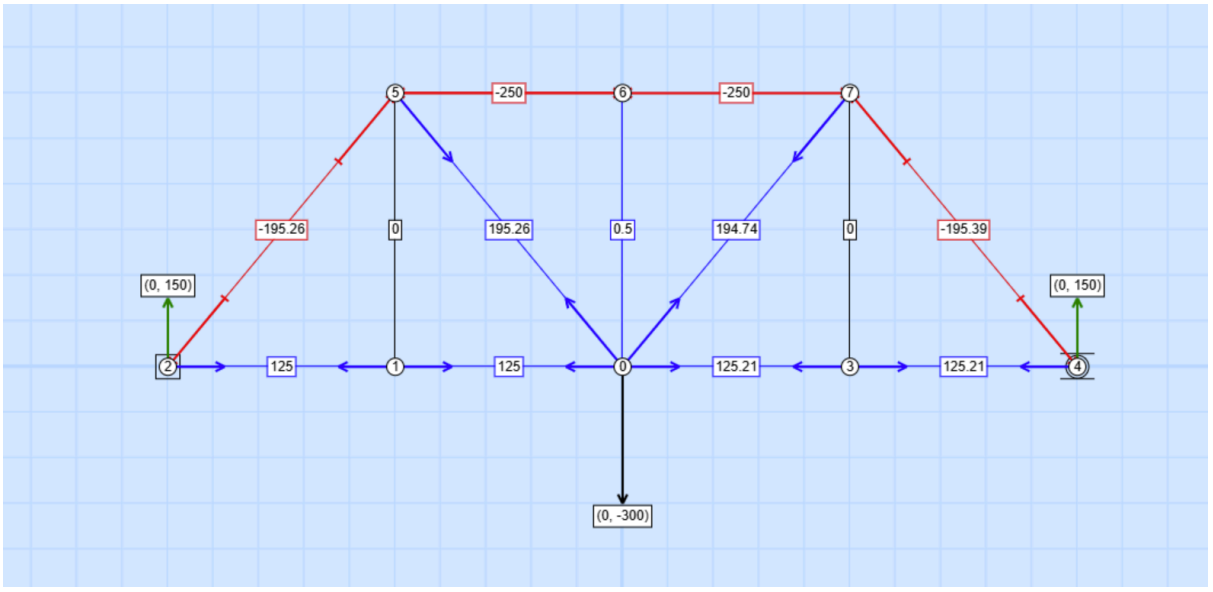
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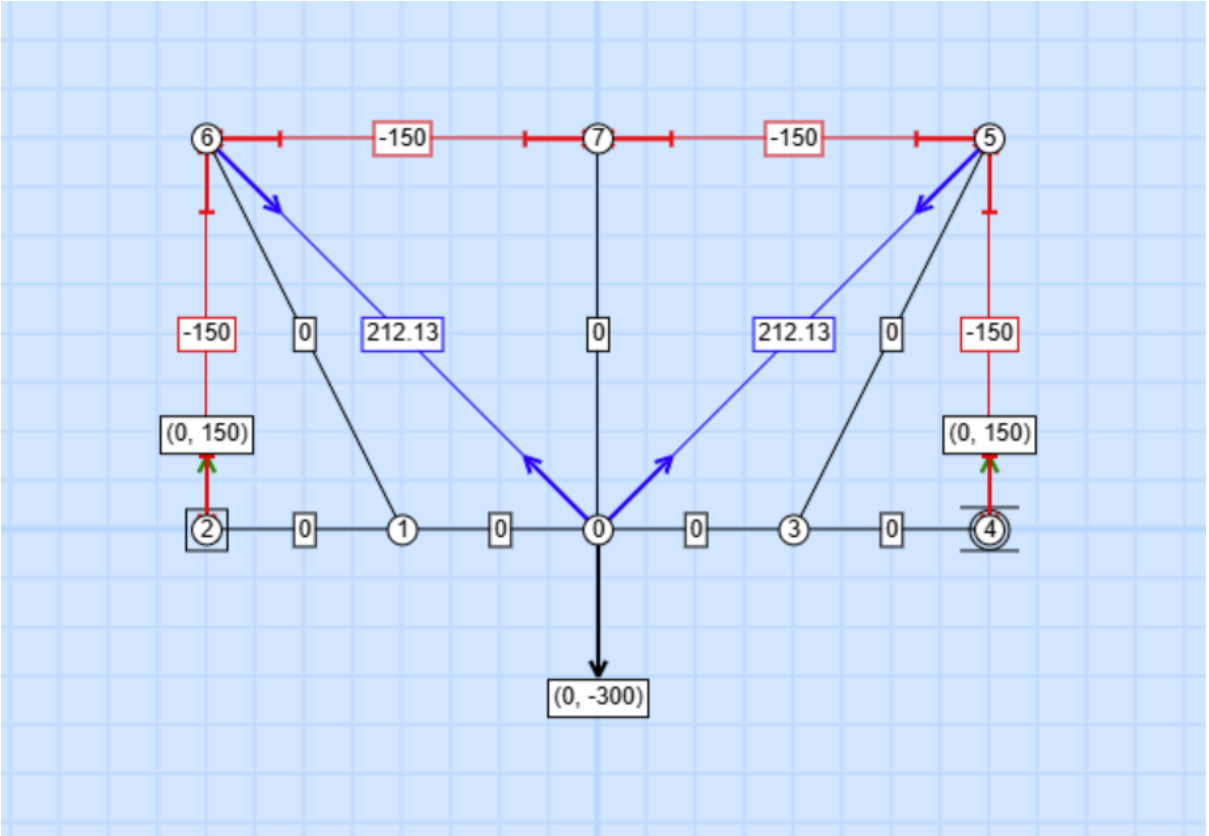

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